Two coprime prime-factor sources beyond every primorial

OEIS A002110 — 12 November 2026

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Abstract. If P_n is the product of the first n primes, $n \geq 2$, the two integers $P_n - 1$ and $P_n + 1$ each have a prime divisor exceeding p_n. The divisors can be chosen distinct. This elementary assertion is a consequence of the primorial definition, not a claim of new mathematics or a proof that either neighboring integer is itself prime.

Sequence and scope. OEIS A002110 defines $a(0) = 1$ and $a(n) = P_n = \prod_{j=1}^{n} p_j$ for $n \geq 1$, where p_j is the jth prime [1]. The statements below apply to $n \geq 2$; the case $n = 1$ is excluded because $P_1 - 1 = 1$ has no prime divisor.

Theorem 1. *For every integer $n \geq 2$ there are distinct primes q_n^-, q_n^+ satisfying*

$$q_n^- \mid (P_n - 1), \qquad q_n^+ \mid (P_n + 1), \qquad q_n^- > p_n, \quad q_n^+ > p_n.$$

Indeed, all prime divisors of either $P_n - 1$ or $P_n + 1$ exceed p_n.

Proof. Let r be any prime with $r \leq p_n$. Then r occurs among $p_1, \ldots, p_n$, so $P_n \equiv 0 \pmod{r}$. Consequently $P_n - 1 \equiv -1 \pmod{r}$ and $P_n + 1 \equiv 1 \pmod{r}$, and r divides neither integer. Since $n \geq 2$, $P_n \geq 6$, and both $P_n - 1$ and $P_n + 1$ exceed 1; each therefore has a prime divisor, necessarily greater than p_n. Moreover P_n is even, so both neighboring numbers are odd. Their greatest common divisor divides their difference 2 and is odd, hence

$$\gcd(P_n - 1, P_n + 1) = 1.$$

Thus no prime can divide both numbers and $q_n^- \neq q_n^+$, as claimed. □

Example. $P_3 = 30$, so $P_3 - 1 = 29$ and $P_3 + 1 = 31$ have distinct prime factors exceeding $p_3 = 5$.

Priority and verification note. This is a routine Euclidean divisibility lemma about OEIS A002110. It does *not* establish the infinitude of prime values among the numbers $P_n \pm 1$, which is a separate open issue [2]. The proof and small cases were checked by AI; no independent human referee or novelty certification is claimed.

References. [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, A002110, `https://oeis.org/A002110` (accessed 12 Nov. 2026). [2] H. Dubner, "Factorial and primorial primes," *J. Recreational Mathematics* 19 (1987), 197–203; linked from `https://oeis.org/A006794`.